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Spatial Asymmetry in Autonomous Vehicle Efficiency Gains for Urban Commuting: A City-Wide Microscopic Simulation Study in Beijing

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Abstract

Autonomous Vehicles (AVs) have been widely recognized as a promising solution to urban commuting congestion. However, quantitative evidence based on city-scale simulations of complete road networks in megacities remains limited. This study uses the complete urban road network of Beijing to investigate the influence of autonomous driving on commuting efficiency. Eleven autonomous vehicle penetration scenarios ranging from 0% to 100% at 10% intervals are established within the Simulation of Urban MObility (SUMO) microscopic traffic simulation platform. Human-driven vehicles are modeled using the Krauss car-following model, whereas autonomous vehicles are represented by the Cooperative Adaptive Cruise Control (CACC) model. The vehicle behavioral parameters are literature-based, adopted from published studies and open test data rather than calibrated against empirical Beijing traffic data, while the road network and commuting demand are constructed from Beijing-specific OpenStreetMap and mobile-signaling data. The simulation results reveal three major findings. First, autonomous driving exhibits a gradual efficiency transition over an approximate penetration range of 30% to 50% (identified qualitatively from the simulation trend rather than by a formal statistical change-point estimate). Below this threshold, behavioral heterogeneity between autonomous and human-driven vehicles intensifies traffic flow instability, whereas above it, the cooperative control capability of CACC becomes dominant and substantially improves overall network performance. Second, under full autonomous vehicle penetration, the city-wide average commuting speed increases from 7.20 m/s to 8.27 m/s, representing a 15% gain in the trip-weighted mean commuting speed (distinct from the 16% gain in the flow-weighted network speed reported in the Results), while the mean in-network simulated travel time per completed trip decreases from 561 s to 270 s. This travel-time value is an operational in-network measure and is not directly comparable to a full perceived door-to-door commute. Third, the efficiency benefits of autonomous driving display significant spatial heterogeneity. Speed improvements reach 16% to 20% on expressways and radial commuting corridors but remain between 4% and 8% on urban arterial roads. These findings indicate that the potential efficiency gains associated with autonomous driving, estimated here under fixed commuting demand and therefore as an upper bound, are constrained by the spatial characteristics of the road network. The results provide quantitative evidence supporting priority deployment of autonomous vehicles on expressways and major commuting corridors in megacities.

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Keywords: AV penetration rate; SUMO traffic simulation; urban commuting; mixed traffic flow; spatial asymmetry

1. Introduction

This section introduces the motivation for the study, identifies the research gap in city-scale evaluations of autonomous vehicle effects on commuting, and states the objectives and contributions. It is organized into two parts: the research gaps and motivations, and the specific research contributions of this work.

1.1. Research Gaps and Motivations

Commuting inefficiency remains one of the most persistent challenges facing transportation systems in megacities. In 2023, only 58% of commuters in Beijing completed a one-way commute within 45 min, whereas 28% required more than 60 min, indicating one of the lowest levels of commuting performance among major Chinese megacities [1,2]. The average one-way commuting distance in Beijing reached 11.6 km, approximately 20% longer than that in Shanghai and 50% longer than that in Shenzhen. In addition, long-distance commuters traveling more than 25 km accounted for 12% of the commuting population [3,4]. This high-intensity commuting pattern is primarily driven by a persistent jobs-housing spatial mismatch, characterized by the concentration of employment opportunities in central urban districts and the continued outward expansion of residential areas. The resulting structural imbalance places sustained pressure on the urban backbone road network, particularly along major expressways and radial commuting corridors, leading to recurrent congestion bottlenecks during peak periods [5–7].

Against this background, Autonomous Vehicle (AV) technology has emerged as a promising approach to mitigating urban commuting congestion. By reducing driver reaction time, maintaining shorter and more stable vehicle headways, and enabling cooperative vehicle control through Cooperative Adaptive Cruise Control (CACC), autonomous vehicles have the potential to increase roadway capacity, enhance traffic flow stability, and improve overall network efficiency [8–11]. Throughout this study, the AV penetration rate is defined as the proportion of autonomous vehicles within the total simulated vehicle fleet; for example, a 40% penetration rate denotes that 40% of the loaded vehicles are autonomous and the remaining 60% are human-driven. Meanwhile, the rapid advancement of autonomous driving policies and pilot programs in China has further strengthened the practical significance of evaluating their potential impacts. National strategic initiatives have identified autonomous driving as a key direction for intelligent transportation development, while Beijing has continuously expanded large-scale autonomous vehicle testing and demonstration zones, providing a favorable policy context for this study.

Despite substantial progress in autonomous vehicle research, several important knowledge gaps remain. First, existing simulation studies have predominantly focused on isolated road segments, corridors, or simplified synthetic networks, whereas city-scale microscopic simulations based on complete urban road networks remain scarce. Second, previous studies have primarily evaluated the impacts of autonomous vehicles using supply-side traffic performance indicators, such as roadway capacity and traffic throughput, rather than commuter-oriented measures that directly reflect travel experience, including commuting time and travel speed. Furthermore, the spatial heterogeneity of autonomous vehicle benefits across different road hierarchy levels has not been systematically quantified at the city scale.

To address these gaps, this study investigates the following research questions:

RQ1. Does autonomous vehicle penetration exhibit a critical threshold beyond which city-wide commuting efficiency improves substantially?
RQ2. How do city-wide commuting time and travel speed respond to full autonomous vehicle penetration?
RQ3. How are the efficiency gains associated with autonomous vehicles distributed across different road hierarchy levels, and what mechanisms drive their spatial heterogeneity?

1.2. Research Contributions

To address the research gaps identified above, this study develops a city-scale microscopic traffic simulation of the complete Beijing Road network under eleven autonomous vehicle penetration scenarios, evaluating both commuter-level travel experience and network-level spatial performance. Building on this framework, this study contributes to the literature in three aspects:

(1) This paper integrates city-scale microscopic traffic simulation with commuter-oriented performance evaluation. Using the complete urban road network of Beijing, this study evaluates autonomous vehicle impacts through commuter travel time and travel speed, providing a city-scale assessment of door-to-door commuting performance while capturing network-wide traffic interactions.
(2) This paper quantifies the nonlinear relationship between autonomous vehicle penetration and commuting efficiency. The results identify a transition region near 40% penetration, below which mixed traffic limits efficiency gains, whereas above this level cooperative vehicle control substantially improves network performance. This provides quantitative evidence for threshold behavior within a real-world metropolitan road network.
(3) This paper reveals the spatial heterogeneity of autonomous vehicle benefits across different road hierarchy levels and proposes a road network structure amplification framework to explain the underlying mechanism. The results show that autonomous vehicles generate substantially greater efficiency gains on higher-order roads, indicating that their operational benefits depend strongly on road network structure rather than being spatially uniform.

This study advances the understanding of autonomous vehicle impacts on urban transportation by integrating city-scale microscopic simulation with commuter-oriented performance evaluation, quantifying the nonlinear relationship between autonomous vehicle penetration and commuting efficiency, explaining the spatial heterogeneity of efficiency gains across the urban road network. These findings provide quantitative evidence to support transportation planning and autonomous vehicle deployment strategies in megacities.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the research methodology. Section 4 describes the study area and simulation framework. Section 5 presents the simulation results. Section 6 discusses the underlying mechanisms and planning implications. Finally, Section 7 concludes the paper and outlines future research directions.

2. Literature Review

This section reviews prior work most relevant to the present study. It is organized into two parts: studies on autonomous driving and traffic operations, including car-following behavior, mixed-traffic dynamics, and network capacity; and studies on autonomous driving and commuting behavior at the urban scale. The review situates the present city-scale, spatially differentiated analysis relative to existing segment-level and network-level work.

2.1. Literature Review on Autonomous Driving and Traffic Operations

The development of autonomous driving technology is profoundly transforming the operational characteristics of transportation systems, exerting significant influence on traffic-flow efficiency, operational stability, safety, and environmental performance [12]. Quantitative research on AV traffic-flow impacts relies primarily on microscopic traffic simulation. For Human-driven Vehicle (HV) modeling, the Krauss car-following model is widely employed in platforms such as Simulation of Urban MObility (SUMO) because of its ability to simulate driver operational variability through a stochastic deceleration mechanism [13–16]. For AV modeling, ACC and CACC car-following models are the two most widely adopted paradigms [8,9,11]. Existing research shows that AVs possess shorter reaction times, more stable headways, and stronger car-following consistency, theoretically offering significant potential to enhance traffic-flow stability [17].

The impacts of autonomous vehicles on traffic performance are strongly influenced by market penetration and exhibit pronounced nonlinear characteristics. Existing studies suggest that, under low penetration rates, typically below 30% to 40%, behavioral heterogeneity between autonomous and human-driven vehicles may reduce traffic flow stability under mixed traffic conditions [18,19]. At the urban network scale, reported efficiency gains are generally smaller than those observed on freeways, indicating that complex urban traffic environments may constrain the operational benefits of autonomous driving [20,21]. A comparison of representative studies is provided in Table 1. This nonlinear response can be explained by the interaction between heterogeneous driving behaviors. At low penetration rates, differences in car-following characteristics, including reaction time and desired headway, introduce local traffic disturbances that limit system-level efficiency improvements. As penetration increases, the formation of stable cooperative vehicle platoons enables CACC to enhance traffic flow stability and gradually improve network performance [11].

Table 1. Comparison of major simulation studies on AV traffic flow impacts.

Literature	Method	Key Finding	Limitation
[18] Ntousakis et al. (2015)	CACC penetration simulation, freeway	Capacity improvement beyond 20% penetration	Single segment; not an urban network
[22] Ghiasi et al. (2017)	Analytical model, mixed-traffic capacity	100% penetration can raise capacity by 80%	Theoretical upper bound; excludes urban intersections
[20] Aria et al. (2016)	SUMO urban network simulation	Urban network benefits lower than freeway	Small synthetic network; not a real city
This study	SUMO full network, Beijing	40% threshold + spatial asymmetry	Fixed OD; excludes induced demand

2.2. Literature Review on Autonomous Driving and Commuting Behavior

At the individual level, autonomous vehicles may transform in-vehicle travel time from a perceived travel cost into productive time, thereby reducing the disutility of commuting and increasing individuals' willingness to undertake longer trips [23]. However, reductions in generalized travel costs may also induce additional travel demand. According to induced-demand theory, these behavioral responses may partially offset the long-term efficiency gains associated with autonomous driving [24,25]. To isolate the direct effects of autonomous vehicle penetration on network performance, this study assumes a fixed origin-destination demand matrix and therefore does not account for induced-demand effects.

From a theoretical perspective, the spatial heterogeneity of autonomous vehicle benefits can be interpreted through the concepts of symmetry and symmetry breaking in network systems. In an idealized topologically homogeneous road network, where all links share identical geometric and operational characteristics, autonomous vehicle deployment would be expected to produce spatially uniform, or symmetric, improvements in

traffic performance across the entire network. Real urban road networks, however, are hierarchically structured, comprising expressways, arterials, and local streets that differ systematically in signal density, intersection spacing, and traffic composition. This structural heterogeneity acts as a symmetry-breaking factor: it causes the operational benefits of autonomous vehicles, which are governed by uniform car-following and cooperative control mechanisms, to be unevenly expressed across the network depending on local roadway conditions. Although several studies have examined how road hierarchy moderates AV performance gains [20,21]. few have explicitly framed this phenomenon as a network-level symmetry-breaking process or attempted to quantify its degree. This study addresses this gap by both empirically characterizing and formally quantifying the spatial asymmetry of AV efficiency gains across the Beijing Road network.

Overall, the existing literature exhibits three important limitations. First, most studies focus on freeways, corridors, or simplified synthetic networks, whereas city-scale microscopic simulations based on complete road networks in real megacities remain scarce. Second, previous research has primarily relied on supply-side traffic performance measures, with limited attention to commuter-oriented outcomes such as travel time and travel speed. Third, empirical evidence from Asian megacities remains limited. This study addresses these gaps through city-scale microscopic simulation of the complete urban road network in Beijing.

3. Methodology

This section describes the modeling approach used to evaluate autonomous vehicle penetration effects. It first presents the overall research framework, then the SUMO microscopic simulation environment, and finally the vehicle parameter settings and the Krauss and CACC car-following models used to represent human-driven and autonomous vehicles.

3.1. Research Framework

The methodological framework of this study is illustrated in Figure 1 and consists of four components: data preparation, microscopic traffic simulation, scenario design, and performance evaluation. The complete urban road network of Beijing serves as the simulation environment, and the impacts of autonomous vehicle deployment are evaluated from two complementary perspectives: commuter-level performance, measured by trip-weighted travel time and travel speed for actual commuting trips, and network-level performance, measured by link traffic load and the unweighted average speed across all network links. Because the network-level indicator includes low-volume or lightly traveled links that are largely unaffected by AV deployment, its magnitude of change is not directly comparable to the trip-weighted commuter-level speed improvement reported in Section 5.1.2. A total of eleven scenarios with autonomous vehicle penetration rates ranging from 0% to 100% are designed to quantify the influence of increasing market penetration. To isolate the direct effects of autonomous vehicle penetration on traffic performance, a fixed commuting origin-destination demand matrix is assumed throughout the simulations, and induced-demand effects are not considered.

3.2. SUMO Simulation Framework

This study employs SUMO (version 1.24.0), an open-source microscopic traffic simulation platform widely used in transportation research [13,14,26]. SUMO provides native support for CACC models, efficient Python-based scenario management through the *TraCI* interface, and integrated network and demand preprocessing tools, making it well suited for large-scale autonomous vehicle simulation. The complete simulation workflow is illustrated in Figure 2.

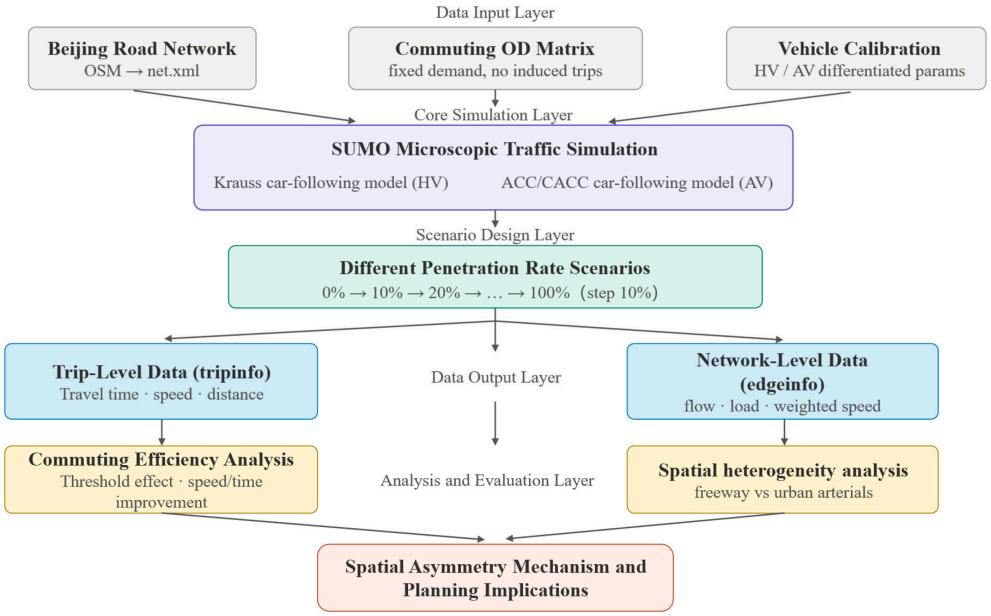

Figure 1. Overall methodological framework of this study.

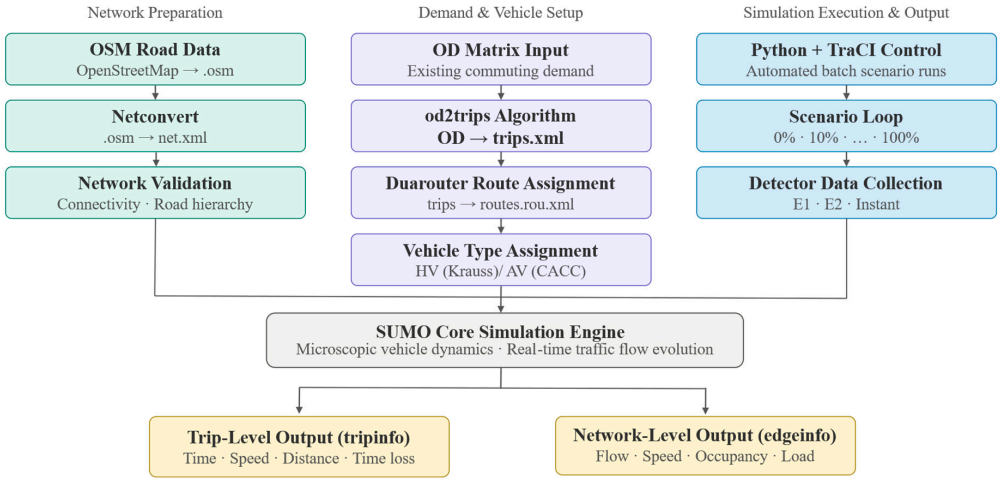

Figure 2. SUMO simulation workflow.

3.3. Vehicle Parameter Calibration and Car-Following Models

3.3.1. Vehicle Parameter Calibration

Based on published literature and open test data, differentiated parameter settings are applied to HVs and AVs, and the results are shown in Table 2.

Parameter sourcing note: the human-driven vehicle parameters (maximum acceleration, comfortable deceleration, and safe time headway) are adopted from the Krauss-model settings reported by Kesting et al. (2008) [17] and the SUMO default calibration ranges [15], and the autonomous vehicle parameters are adopted from the CACC implementation and field experiments reported by Milanés and Shladover (2014) [9]. These values are literature-based rather than calibrated against Beijing field data; the maximum acceleration and comfortable deceleration entries in Table 2 correspond to the SUMO accel and decel attributes and fall within the ranges reported in these sources. No formal calibration or optimization procedure was performed, and the uncertainty associated with the model parameters constitutes a limitation of the present study.

Table 2. Vehicle parameter calibration for human-driven and autonomous vehicles.

Vehicle Type	Max. Acceleration (m/s^2)	Comfortable Deceleration (m/s^2)	Safe Time Headway (s)
Human-driven (HV)	1.02	5.04	1.25
Autonomous (AV)	2.53	6.33	0.78

3.3.2. Human-Driven Vehicle Modeling Using the Krauss Car-Following Model

HVs adopt the Krauss model, and safe speed and desired speed are computed by Equations (1) and (2), respectively. The safe speed $v_s(t)$ of an HV at time t is:

$$v_s(t) = \sqrt{(-\tau \cdot b)^2 + v_l(t-1)^2 + 2 \cdot b \cdot g_l(t-1)} \quad (1)$$

where $v_s(t)$ denotes the safe speed of the following vehicle at time t (m/s); τ is the reaction time of the following vehicle (s), typically ranging from 1.0 to 1.5 s for human-driven vehicles; b is the maximum deceleration (m/s^2); $v_l(t-1)$ denotes the speed of the leading vehicle at time $t-1$ (m/s); and $g_l(t-1)$ represents the net gap between the leading and following vehicles at time $t-1$ (m).

Desired Speed Equation. The desired speed $v_d(t)$ is the minimum of the following three terms, ensuring both safety and kinematic constraints are satisfied:

$$v_d(t) = min\{v_s(t), v(t-1) + a, v_{max}\} \quad (2)$$

where $v_d(t)$ denotes the desired speed of the vehicle at time t (m/s), $v(t-1)$ denotes the vehicle speed at time $t-1$ (m/s), a denotes the maximum acceleration (m/s^2), and v_{max} denotes the maximum allowable speed of the vehicle (m/s). The model incorporates a stochastic deceleration component to represent the inherent variability in human driving behavior.

3.3.3. CACC Car-Following Model for Autonomous Vehicles

AVs adopt the CACC model, automatically switching among three control modes based on the current time gap.

Mode (i): Speed/Cruise Control Mode. Activated when the time gap exceeds 2 s; the vehicle enters free-cruise mode. The acceleration control law is:

$$\alpha_{i,k+1} = k_1(v_d - v_{i,k}), k_1 > 0 \quad (3)$$

where $\alpha_{i,k+1}$ denotes the acceleration of vehicle i at time step $k+1$ (m/s^2), v_d denotes the target cruising speed (m/s), $v_{i,k}$ denotes the speed of vehicle i at time step k (m/s), and k_1 denotes the control gain, which is set to 0.4 s^{-1}.

Mode (ii): Gap Control Mode. Activated when the gap error and speed error are simultaneously less than 0.2 m and 0.1 m/s, respectively. The speed update law is:

$$v_{i,k+1} = v_{i,k} + k_2 e_{i,k} + k_3 \dot{e}_{i,k}, k_2, k_3 > 0 \quad (4)$$

$$e_{i,k} = x_{i-1,k} - x_{i,k} - t_d v_{i,k} \quad (5)$$

$$\dot{e}_{i,k} = v_{i-1,k} - v_{i,k} \quad (6)$$

where t_d denotes the desired time gap of the Cooperative Adaptive Cruise Control controller (s), and k_2 and k_3 denote the longitudinal control gains, with values of 0.45 s^{-2} and 0.25 s^{-2}, respectively.

Mode (iii): Gap-Closing Control Mode. Activated when the time gap falls below 1.5 s. This mode adjusts the parameters of the gap control mode; the optimal control gains are:

$$k_2 = 0.01\,\mathrm{s}^{-2} \tag{7}$$

$$k_3 = 1.6\,\mathrm{s}^{-2} \tag{8}$$

Compared with Mode (ii), in which $k_2 = 0.45\,\mathrm{s}^{-2}$ and $k_3 = 0.25\,\mathrm{s}^{-2}$, Mode (iii) adopts a substantially higher speed-error gain, increasing k_3 from 0.25 to 1.60 s^{-2}, thereby enhancing the responsiveness of the controller to speed deviations. Meanwhile, the gap-error gain k_2 is reduced to 0.01 s^{-2}, which moderates the spacing-control response and improves control stability by mitigating oscillatory behavior. The CACC gains used above ($k_1 = 0.4\,\mathrm{s}^{-1}$ in Mode (i); $k_2 = 0.45\,\mathrm{s}^{-2}$ and $k_3 = 0.25\,\mathrm{s}^{-2}$ in Mode (ii); $k_2 = 0.01\,\mathrm{s}^{-2}$ and $k_3 = 016\,\mathrm{s}^{-2}$ in Mode (iii)) are adopted from the CACC controller design and field experiments of Milanés and Shladover (2014) [9] and the SUMO CACC implementation documentation; the gains are reported in the units shown and were used at these values in all scenarios.

A comparison of key parameters for the two car-following models is shown in Table 3.

Table 3. Qualitative comparison of Krauss and CACC model characteristics and their effects on capacity.

Parameter	Krauss (HV)	CACC (AV)	Impact on Capacity
accel is maximum acceleration (m/s^2)	Conservative (see calibrated value in Table 2)	Larger (see calibrated value in Table 2)	Affects start-up and recovery
decel is maximum deceleration (m/s^2)	Comfort-oriented, lower magnitude	Can approach the safety limit (see calibrated value in Table 2)	Affects stable car-following
τ is driver/controller reaction time (s), distinct from the safe time headway calibrated in Table 2	Implicit and comparatively large	Explicit, small	Directly determines capacity
min Gap is minimum static gap (m)	Small	Often larger	Low-speed safety

To transform macroscopic travel demand into microscopic vehicle movements, the SUMO *od2trips* module is employed to convert origin-destination demand into individual vehicle trips, with autonomous vehicles and human-driven vehicles represented as distinct vehicle types. Behavioral differences between the two vehicle classes are specified through car-following parameters, including reaction time, acceleration and deceleration capabilities, desired headway, and driving stability, thereby capturing the distinct operational characteristics of mixed traffic. Vehicle routes are subsequently generated using the *duarouter* algorithm, which converts trip demand into executable vehicle trajectories and establishes the linkage between macroscopic travel demand and microscopic traffic flow.

4. Case Study and Network Construction

4.1. Network Construction and Traffic Demand Loading

To evaluate the impacts of autonomous vehicle deployment on urban traffic systems and commuting performance, this study adopts the complete road network of Beijing as the simulation network and develops a city-scale microscopic traffic simulation model. A series of autonomous vehicle penetration scenarios is constructed to systematically assess changes in network-wide traffic performance and commuter travel efficiency under different levels of autonomous vehicle adoption. The road network of Beijing used in this study is shown in Figure 3.

During the data preparation stage, the complete road network of Beijing was obtained from the OpenStreetMap platform and converted into a SUMO-compatible *net.xml* network using the *Netconvert* module, thereby reconstructing the urban road network for microscopic traffic simulation. Travel demand was derived from anonymized mobile signaling data to

generate the baseline origin-destination matrix, while induced-demand effects associated with autonomous vehicle deployment were not considered. The origin-destination matrix was constructed by identifying user stay locations from base-station connection records, inferring home and workplace locations according to dwell duration and temporal activity patterns, and extracting home-to-work commuting trips during the morning peak period from 07:00 to 09:00. Finally, traffic analysis zones were spatially matched to the road network to enable accurate loading of travel demand, thereby supporting city-scale traffic assignment and microscopic traffic simulation.

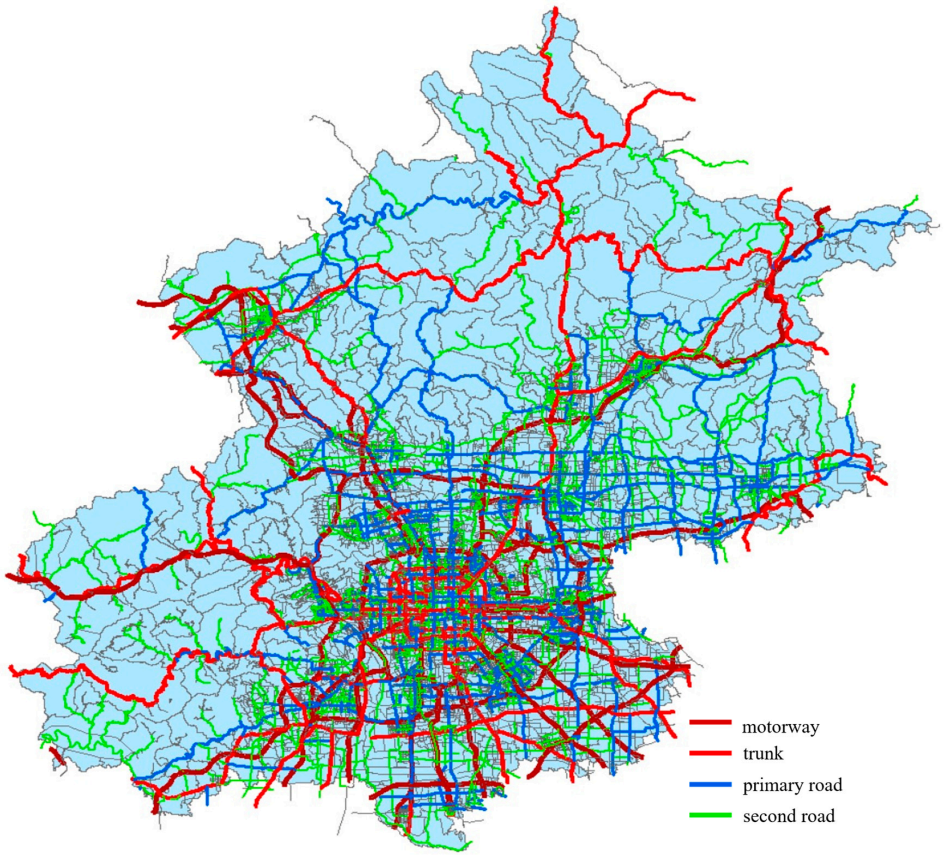

Figure 3. Beijing road network used in this study.

For readers unfamiliar with OpenStreetMap (OSM) classification, the OSM highway tags used in Figure 3 map to functional road classes as follows: motorway denotes grade-separated expressways (typical speed limit 80 to 120 km/h); trunk denotes major through-roads and ring expressway segments (60 to 80 km/h); primary denotes primary urban arterials (50 to 60 km/h); secondary denotes secondary arterials and important collectors (40 to 50 km/h); and tertiary and residential denote local collector and access roads (30 to 40 km/h). These classes correspond to the road-hierarchy categories used in the spatial-asymmetry analysis in Section 5.2.

The complete road network of Beijing is obtained from OpenStreetMap (OSM) and converted into a SUMO-compatible *net.xml* network using the *Netconvert* module. The baseline origin-destination matrix is transformed into microscopic vehicle trips using *od2trips*, followed by route assignment with *Duarouter*, while autonomous vehicles and human-driven vehicles are represented as distinct vehicle types through separate behavioral parameter settings. Three types of detectors, summarized in Table 4, are deployed to collect traffic information at multiple spatial scales, ranging from individual vehicle movements to link-level operations and network-wide performance. The simulation outputs include

both individual vehicle trajectories and travel information through *tripinfo*, and network operating conditions through *edgeinfo*.

Table 4. Detector types and functions in the SUMO simulation.

No.	Detector Type	Primary Data Collected
1	Aggregate inductive-loop detector (E1)	Vehicle count, mean speed, and other flow parameters within the counting interval
2	Disaggregate inductive-loop detector (Instant)	Passage time and passage speed for each individual vehicle
3	Area detector (E2)	Mean travel time, mean speed, mean delay, etc., within the zone

4.2. Scenario Design

Based on the SUMO simulation framework, eleven autonomous vehicle penetration scenarios are established, with penetration rates ranging from 0% to 100% at 10% intervals, as summarized in Table 5. All scenarios are executed in batch through the Python-based *TraCI* interface, while traffic demand, network configuration, and simulation parameters are kept identical to isolate the effects of autonomous vehicle penetration.

Table 5. Autonomous vehicle penetration rate scenario design.

Scenario	AV Penetration	AV Share	Representative Stage
S0	0%	0/10	Baseline: all human-driven
S1	10%	1/10	Early AV pilot
S2	20%	2/10	Small-scale demonstration
S3	30%	3/10	Low-penetration transition
S4	40%	4/10	Mid-range penetration scenario
S5	50%	5/10	Mixed-traffic equilibrium
S6	60%	6/10	AV-dominant stage
S7	70%	7/10	High-penetration diffusion
S8	80%	8/10	Deep penetration
S9	90%	9/10	Near-full autonomous driving
S10	100%	10/10	Full autonomous driving

5. Results

5.1. City-Wide Commuting Efficiency Change Patterns

5.1.1. Network Operational Indicator Analysis

Based on the simulation framework described above, the study constructs consecutive development scenarios from low to high AV penetration and conducts a comparative analysis of the urban traffic system's operational response during the technology transition and maturity stages. Changes in network operational indicators under different AV penetration rates are shown in Figure 4.

Results show that AV improvements in city-wide network operational efficiency exhibit clear staged and threshold characteristics. During the low-penetration stage, behavioral differences between AVs and HVs cause mixed-traffic operational stability to decline, local link occupancy to rise, and system efficiency to fluctuate temporarily. Once AV penetration exceeds approximately 40%, operational conditions on high-flow roads improve markedly and traffic-flow stability strengthens.

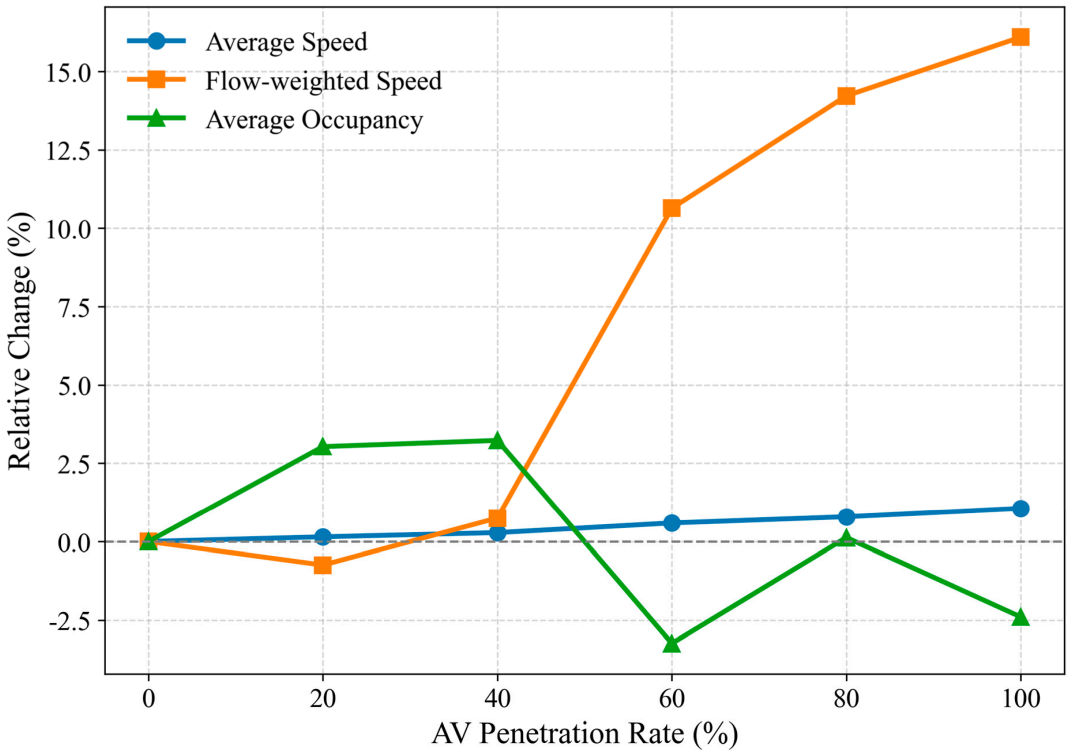

Figure 4. Changes in network operational indicators under different AV penetration.

The occupancy pattern shown in Figure 4 can be explained by the fundamental traffic flow relationship $q = kv$, where detector occupancy serves as a proxy for local traffic density. During the 0–40% penetration range, behavioral heterogeneity between autonomous vehicles and human-driven vehicles increases local traffic disturbances, resulting in a slight increase in occupancy relative to the baseline of approximately 3%. As penetration increases to 40–60%, stable Cooperative Adaptive Cruise Control platoons begin to form, substantially increasing travel speed while reducing average headways. Because traffic density is proportional to q/v, the larger increase in speed relative to throughput leads to a reduction in occupancy, reaching approximately 3.5% below the baseline at 60% penetration. Between 60% and 80%, occupancy gradually returns toward the baseline as the additional network capacity accommodates previously constrained traffic demand. Beyond 80%, Cooperative Adaptive Cruise Control becomes the dominant operating mode, and further improvements in speed and headway reduction again exceed the increase in throughput, producing a second local minimum in occupancy of approximately 2.5% below the baseline. Overall, the observed non-monotonic pattern reflects the evolving balance between travel speed and traffic throughput as the network progressively transitions from mixed traffic to predominantly cooperative traffic operations.

Note that the average speed change shown in Figure 4 is computed as an unweighted average across all network links, including many low-volume local links with limited AV exposure; it therefore understates the improvement experienced by actual commuters, which is reported separately as a trip-weighted measure in Section 5.1.2.

Note on variability: each scenario was simulated once (without multiple random seeds), so run-to-run variability could not be quantified; the curves report the mean indicator value across network links or trips for a single run, and the corresponding trip-level dispersion is discussed in Section 5.1. Quantifying between-run variability through seeded replications is identified as future work.

5.1.2. Overall Improvement in Average Travel Speed

Further analysis from the perspective of residents' commuting characteristics reveals that AV improvements in commuting efficiency derive primarily from optimization of the traffic-operation process rather than from changes in the spatial travel structure. Changes in mean travel speed by vehicle type under different AV penetration rates are shown in Figure 5.

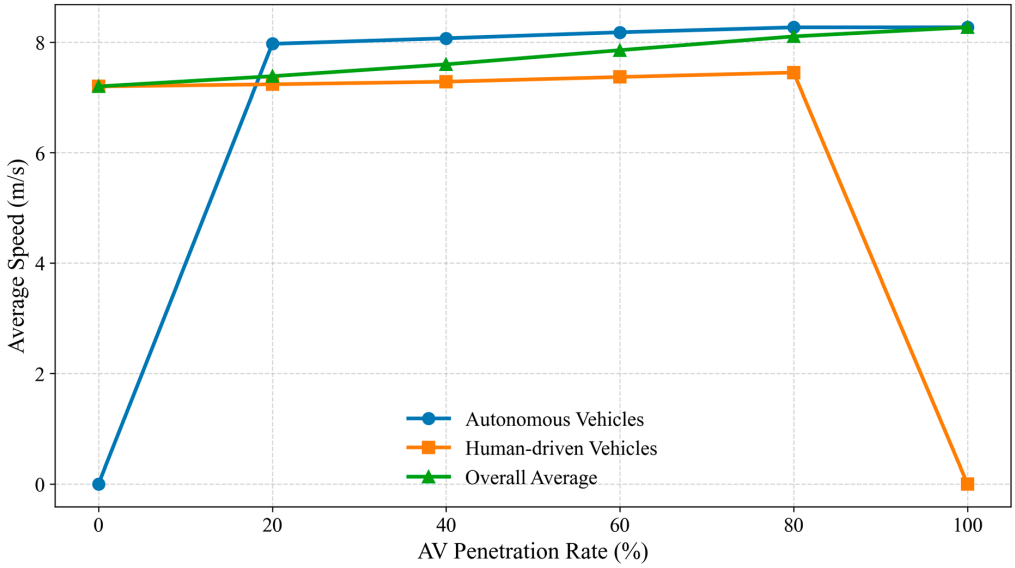

Figure 5. Changes in average travel speed by vehicle type under different AV penetration.

As autonomous vehicle penetration increases from 0% to 100%, the city-wide average travel speed increases from 7.20 to 8.27 m/s, representing an improvement of approximately 15%, while the average simulated travel time decreases from 561 to 270 s, corresponding to a reduction of approximately 52%. In contrast, the average trip distance remains nearly unchanged across all scenarios. These results indicate that autonomous vehicles improve commuting performance primarily by enhancing traffic flow stability, reducing stop-and-go conditions, and increasing the continuity of traffic operations.

A vehicle class is absent at the corresponding penetration extreme (no autonomous vehicles at 0% penetration and no human-driven vehicles at 100%); the zero values shown at these endpoints indicate that the class is not present and should be read as not applicable rather than as a measured speed of zero.

5.1.3. Overall Improvement in Average Travel Time

The quantitative results for each penetration stage are summarized in Tables 6 and 7. Below the 40% penetration threshold, the average travel speed increases only marginally from 7.20 to 7.35 m/s, while the average simulated travel time decreases from 561 to approximately 520 s, corresponding to improvements of less than 10% for both indicators. Beyond the threshold, the flow-weighted average network speed increases progressively with penetration, reaching approximately 16% at full penetration, accompanied by a substantially greater improvement in overall network efficiency relative to the sub-threshold stage.

A vehicle class is absent at the corresponding penetration extreme (no autonomous vehicles at 0% penetration and no human-driven vehicles at 100%); the zero values shown at these endpoints indicate that the class is not present and should be read as not applicable rather than as a measured travel time of zero.

At the city scale, average commuting performance improves progressively with increasing autonomous vehicle penetration, as shown in Table 6. Both average travel speed

and simulated travel time exhibit continuous improvement as the penetration rate increases from 0% to 100%, with substantially greater gains observed at higher penetration rates.

Table 6. Comparison of key commuting efficiency indicators under 0% and 100% AV penetration.

Evaluation Indicator	0% Penetration	100% Penetration	Change
Mean commuting speed (m/s)	7.20	8.27	+1.07 m/s (+14.9%)
Mean simulated travel time (s)	561	270	−291 s (−51.9%)
Mean trip distance	Baseline	Approximately stable	Change < 2% (no significant difference)
Flow-weighted network speed	Baseline	+16%	Gains accelerate beyond 40% threshold, reaching +16% at full penetration

Table 7. Commuting efficiency change characteristics by penetration stage.

Penetration Range	Mean Commuting Speed	Mean Simulated Travel Time	Stage Characteristics
0% (baseline)	7.20 m/s	561 s	All-HV baseline
0–40%	7.20–7.35 m/s	520–561 s	HV/AV behavioral differences introduce instability; minimal efficiency gain
~40% (threshold)	≈7.35 m/s	≈520 s	CACC cooperative effect begins to dominate; network efficiency surges
40–100%	7.35–8.27 m/s	270–520 s	Efficiency improves steadily; gains accelerate at high penetration
100% (final state)	8.27 m/s	270 s	Full-AV optimum; trip distance change < 2%

The city-wide average travel speed increases from 7.20 to 8.27 m/s, representing an improvement of approximately 15%, while the average simulated travel time decreases from 561 to 270 s, corresponding to a reduction of approximately 52%, as shown in Figure 6. By comparison, the average trip distance remains essentially unchanged, varying by less than 2% across all scenarios. The average travel speed of human-driven vehicles also increases with autonomous vehicle penetration, indicating that the cooperative control capability of Cooperative Adaptive Cruise Control improves traffic conditions for both autonomous and human-driven vehicles under mixed traffic. Detailed results for each penetration stage are summarized in Table 7.

5.2. Spatial Differentiation in Network Operational State

5.2.1. Spatial Restructuring of Network Load

Figure 7 illustrates the spatial distribution of network traffic load under 0% autonomous vehicle penetration. High traffic loads are concentrated throughout the central urban area, with persistently congested corridors observed along the East Third Ring Road, Zhongguancun, Beijing West Railway Station, and Anzhen. These areas are characterized by sustained queue formation and heavy traffic demand. Under full AV penetration, as shown in Figure 8, urban traffic pressure undergoes pronounced spatial restructuring: load levels on the aforementioned high-load corridors decline markedly, operational stability on radial expressway channels improves substantially, and congestion patterns shift from continuous queuing to localized fluctuations.

5.2.2. Layered Improvement Characteristics in Operating Speed

Figures 9 and 10 present the spatial distribution of network operating speeds under 0% and 100% autonomous vehicle penetration, respectively. Under the baseline scenario, the

road network operates predominantly at medium-to-low speeds, with pronounced congestion occurring even along major expressways. Under full autonomous vehicle penetration, operating speeds increase across most of the network, with the largest improvements observed on the expressway system, including the Third Ring Road, Fifth Ring Road, Jingtong Expressway, Beijing–Shanghai Expressway, and Airport Expressway. In particular, the Fifth Ring Road shifts from persistent medium-speed operation to a continuous high-speed operating state, with speed improvements of approximately 16–20%. By comparison, urban arterial roads, including Fuxing Road, Fuchengmenwai Avenue, and Guangqumennei Avenue, exhibit more modest improvements of approximately 4–8%, indicating substantially smaller efficiency gains than those observed on expressways.

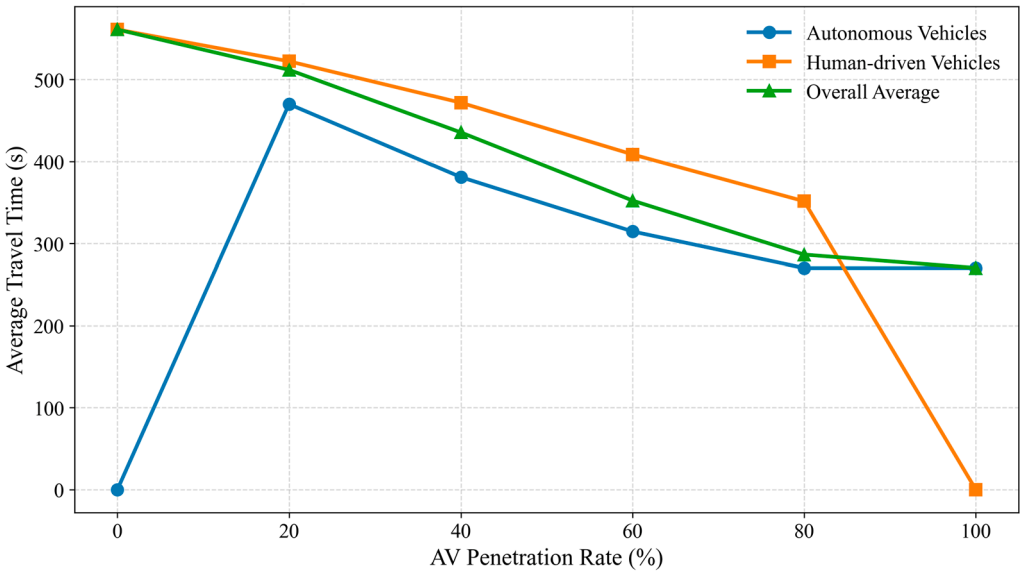

Figure 6. Changes in average travel time by vehicle type under different AV penetration.

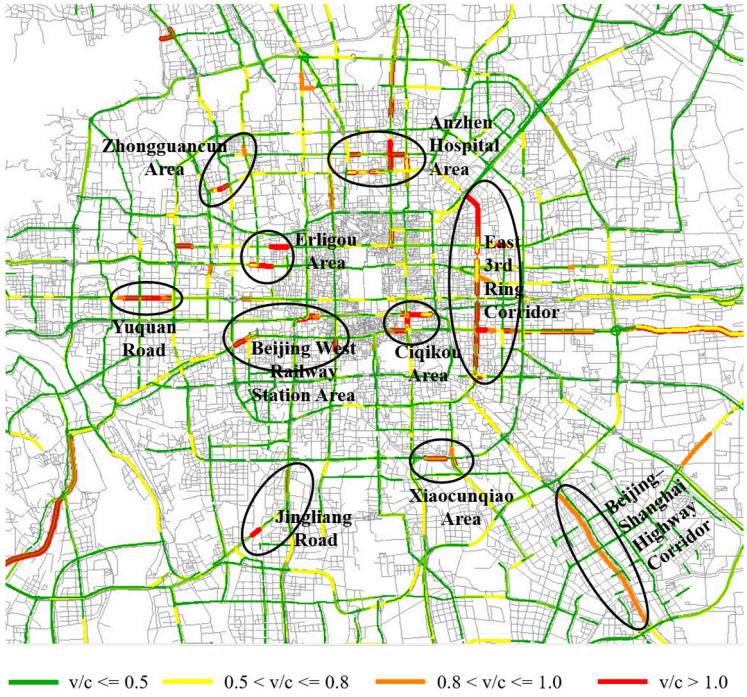

Figure 7. Network load distribution under zero AV penetration.

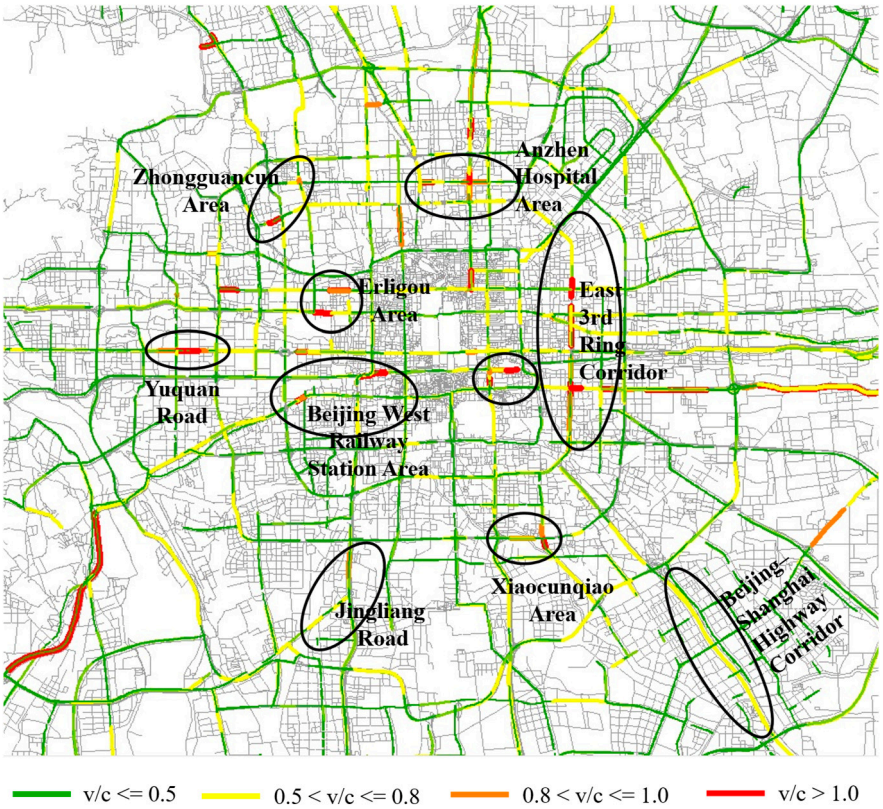

Figure 8. Network load distribution under full AV penetration.

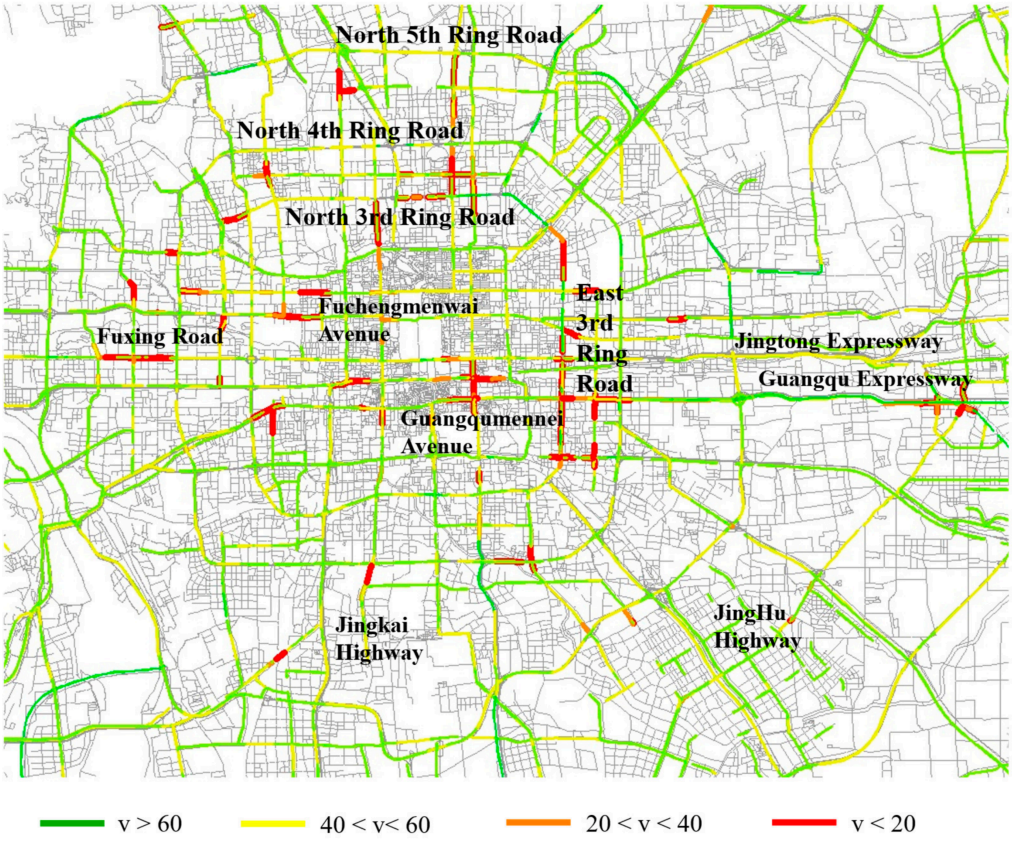

Figure 9. Spatial distribution of network operating speeds under 0% AV penetration.

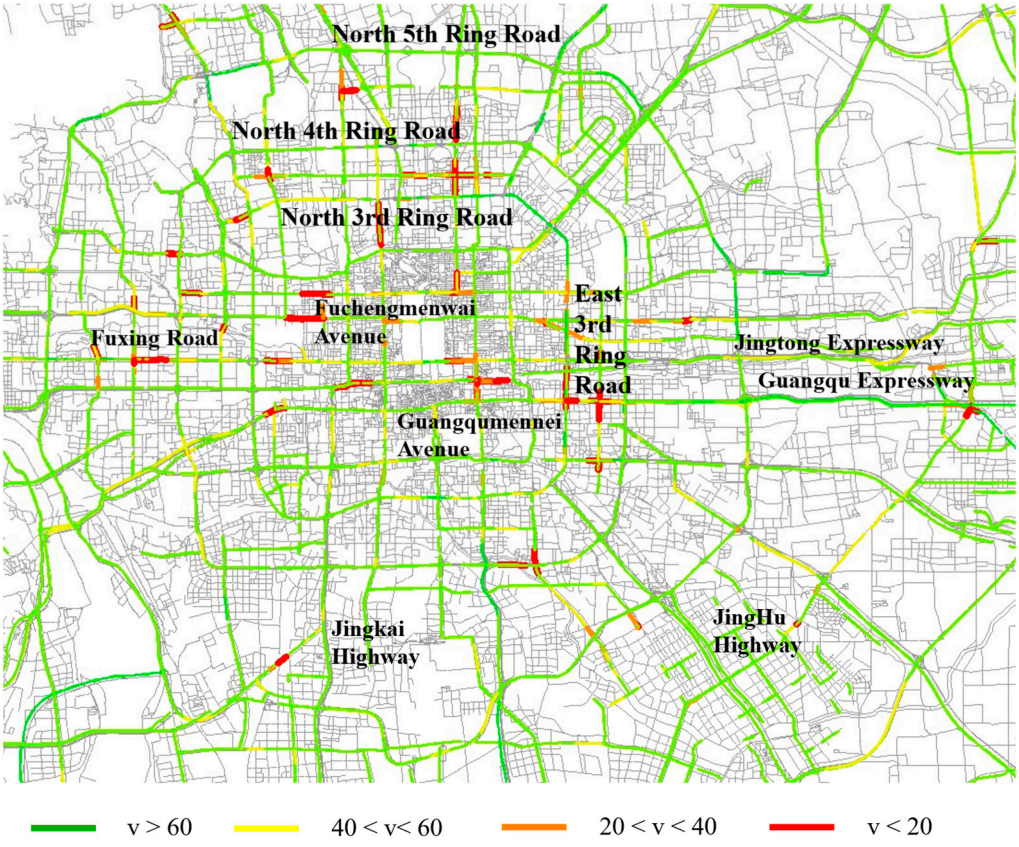

Figure 10. Spatial distribution of network operating speeds under 100% AV penetration.

6. Discussion

6.1. Mechanistic Explanation of the Penetration-Rate Threshold Effect

Within the 0–40% penetration range, heterogeneous car-following behavior between autonomous vehicles and human-driven vehicles introduces additional disturbances under mixed traffic conditions. When human-driven vehicles follow autonomous vehicles, larger safety gaps are typically maintained because of differences in driving behavior and response characteristics. Conversely, when autonomous vehicles follow human-driven vehicles, stochastic fluctuations in human driving behavior limit the effectiveness of Cooperative Adaptive Cruise Control. The cumulative interactions between these heterogeneous vehicle classes increase local traffic flow variability during the low-penetration stage, thereby reducing the overall efficiency gains associated with autonomous vehicle deployment.

Once autonomous vehicle penetration exceeds approximately 40%, the proportion of autonomous vehicles becomes sufficient to support sustained Cooperative Adaptive Cruise Control interactions among neighboring vehicles and the formation of stable autonomous vehicle platoons. From the perspective of traffic flow theory, this pattern is consistent with a transition from an unstable mixed-traffic regime to a more cooperative and stable traffic-flow regime, with approximately 40% penetration representing the corresponding transition threshold. These findings suggest that the efficiency gains associated with autonomous vehicle deployment may increase disproportionately once this threshold is exceeded. From a policy perspective, the results imply that strategies capable of accelerating penetration beyond this transition region may yield greater network-wide benefits than dispersed deployment at relatively low penetration levels. This qualitative transition from an unstable mixed-traffic regime to a cooperative regime is consistent with recent city-scale and mixed-traffic simulation studies. Liu et al. (2024) [27] report that penetration-rate effects on urban network performance are strongly nonlinear and sensitive to demand assumptions, and

Sadid and Antoniou (2024) [28] show that mixed-traffic stability and calibration choices materially affect the estimated benefits of connected and automated vehicles; both findings align with the range-based rather than single-point transition reported here.

6.2. Structural Mechanism Underlying Spatial Asymmetry

The simulation results suggest that the effectiveness of autonomous vehicle deployment depends strongly on roadway operating conditions. Two characteristics appear particularly favorable for realizing autonomous vehicle benefits: sustained traffic flow with limited interruptions and relatively low levels of traffic conflict. Expressways satisfy both conditions by providing uninterrupted traffic flow, homogeneous traffic composition, and limited conflict points, and therefore exhibit the largest efficiency improvements, as shown in Figure 11.

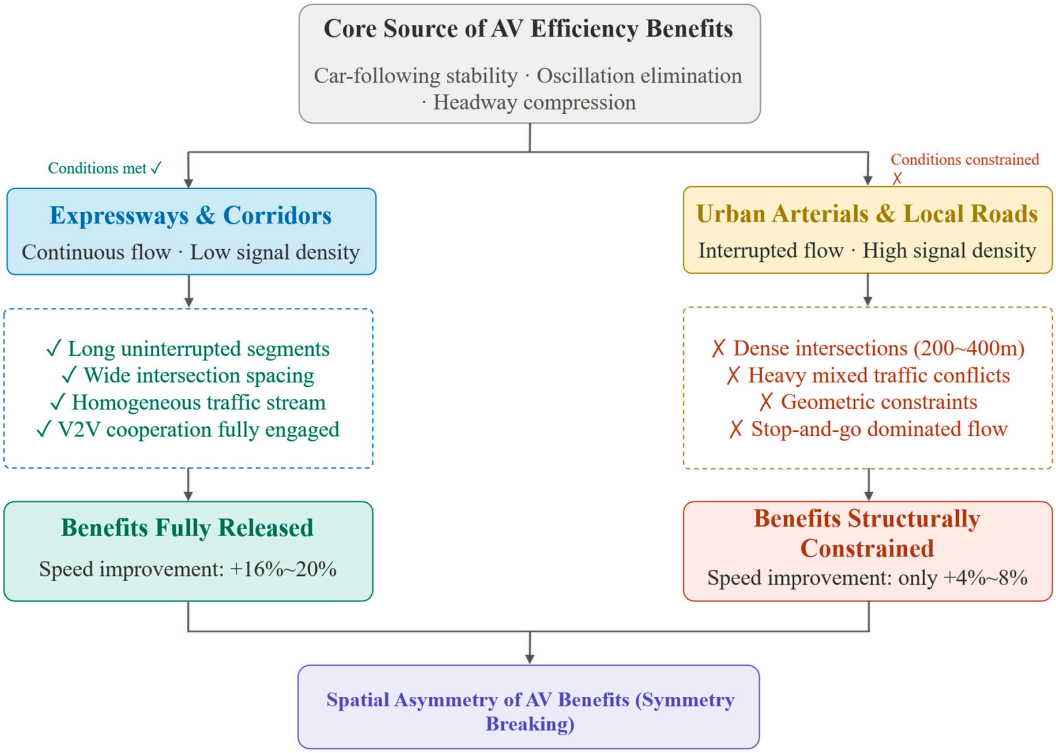

Figure 11. Mechanism underlying spatial asymmetry of AV efficiency gains.

In contrast, urban arterial roads are characterized by closely spaced signalized intersections, typically separated by 200–400 m, requiring vehicles to repeatedly undergo stop-and-go cycles. Frequent interactions with non-motorized traffic and other road users introduce additional uncertainty into vehicle operations, while roadway geometry further limits the effectiveness of Cooperative Adaptive Cruise Control. Under these interrupted traffic conditions, the potential benefits of improved car-following behavior are partially constrained by signal control and intersection operations.

From the perspective of traffic system dynamics, the observed spatial heterogeneity reflects a systematic asymmetry in the distribution of autonomous vehicle benefits across different road classes. Higher-order roads provide operating environments that allow cooperative vehicle control to function more effectively, whereas lower-order urban roads remain constrained by operational complexity. These findings suggest that autonomous vehicle deployment tends to reinforce, rather than eliminate, the performance differences associated with existing road network structures. Consequently, deployment strategies that prioritize higher-order roads may produce greater network-level efficiency gains.

To express this heterogeneity in a more formal, quantitative way, let ΔV_j denote the AV-induced speed improvement for road class j:

$$\Delta V_j = \frac{v_j^{100\%} - v_j^{0\%}}{v_j^{0\%}} \times 100\% \tag{9}$$

where $v_j^{0\%}$ and $v_j^{100\%}$ are the mean operating speeds of road class j under 0% and 100% AV penetration. If AV benefits were spatially uniform (i.e., symmetric), ΔV_j would be the same for every road class. The degree to which this symmetry is broken can be measured using the coefficient of variation of ΔV_j across road classes, referred to here as a Symmetry-Breaking Index (SBI):

$$SBI = \frac{\sigma(\Delta V_j)}{\mu(\Delta V_j)} \tag{10}$$

Based on the midpoint efficiency gains reported in Section 5.2.2, namely 18% for expressways and radial commuting corridors and 6% for urban arterials, the calculated Symmetry Breaking Index is approximately 0.50, indicating a pronounced degree of spatial asymmetry in autonomous vehicle efficiency gains across the Beijing Road network. The proposed Symmetry Breaking Index provides a normalized quantitative measure for characterizing the spatial heterogeneity of autonomous vehicle benefits and is intended to facilitate qualitative comparisons across road network structures and planning scenarios. It should be read as a descriptive summary computed from two representative class-level values (the midpoint efficiency gains for expressways and for arterials) rather than as a validated distributional metric; because it is based on two summary values, it does not capture the full link-level distribution of effects across the city, and the dispersion term is computed as a population standard deviation over the two class values. A recomputation from the full link-level distribution across several well-defined road categories, with an explicit weighting scheme and bootstrap uncertainty intervals, is left to future work. Future research may further improve its applicability by incorporating a more detailed road classification and additional roadway characteristics.

6.3. Implications for Megacity Transportation Planning

The simulation results suggest that the effectiveness of autonomous vehicle deployment may increase substantially once penetration exceeds the identified transition threshold. Before this threshold is reached, deployment strategies that concentrate autonomous vehicles on major commuting corridors, together with investments in vehicle-to-everything infrastructure, may facilitate the formation of cooperative traffic conditions. Complementary policy measures, including vehicle purchase incentives, insurance discounts, and priority access schemes, could further accelerate autonomous vehicle adoption.

The observed spatial heterogeneity also highlights the importance of network-oriented planning. Rather than focusing exclusively on individual roadway segments, planning strategies may benefit from strengthening the hierarchical structure of the road network. Potential measures include improving the continuity of expressway corridors by mitigating bottlenecks, encouraging long-distance travel to use higher-order roads through demand-management policies such as congestion pricing and differentiated parking fees, and prioritizing autonomous vehicle infrastructure upgrades according to road hierarchy. In particular, vehicle-to-everything deployment may be most effective on expressways, whereas arterial roads may benefit more from intersection redesign and signal optimization.

Finally, the present analysis is based on a fixed origin-destination demand matrix and therefore represents an upper-bound estimate of the traffic efficiency gains associated with autonomous vehicle deployment. Because induced travel demand is not explicitly

modeled, the realized benefits may be smaller over the longer term. These findings suggest that autonomous vehicle deployment should be coordinated with demand-management strategies, including congestion pricing and public transport priority, to reduce the potential erosion of efficiency gains resulting from increased travel demand.

6.4. Research Limitations and Future Directions

The principal limitations of this study and corresponding directions for future research are summarized in Table 8. The primary limitation is the assumption of a fixed origin-destination demand matrix. Consequently, the reported efficiency improvements are conditional on an unchanged level of commuting demand and should be interpreted as an upper-bound estimate of the operational benefits associated with autonomous vehicle deployment. Future research could integrate the present microscopic traffic simulation framework with elastic travel demand models and land-use models to develop an integrated evaluation framework that jointly captures the interactions among traffic operations, travel demand, and urban spatial development. Two further limitations concern validation and scope. First, the model is not externally validated: no link-level observed volumes or speeds for the Beijing network were available for the study period, so the results should be read as relative, model-based indications of the direction and structure of AV effects rather than as validated predictions of observed Beijing conditions, and field-data validation using GEH, RMSE and MAPE statistics is left to future work. Second, the simulation represents only private commuting vehicles (human-driven and autonomous) and does not include pedestrians, cyclists, buses, or freight vehicles; the omission of these non-motorized modes, public transit, and freight could affect the magnitude of the estimated effects, particularly on arterials and in mixed-use corridors.

Table 8. Research limitations and future improvement directions.

No.	Limitation Type	Implication	Future Research
1	Fixed origin-destination demand	May overestimate network efficiency gains	Integrate elastic demand and activity-based models
2	[Removed: no residential location-choice model is used in this study] Simplified residential location-choice model	Represents an upper-bound estimate of residential relocation	Develop dynamic land-use–transport models
3	Single peak-period simulation	Does not capture full-day traffic dynamics	Extend to multi-period simulations
4	No vehicle–infrastructure coordination	Benefits of C-V2X are not evaluated	Incorporate roadside sensing and signal coordination
5	Parameter uncertainty	Results depend on literature-based parameter settings (no formal calibration procedure was performed)	Conduct sensitivity analysis and field-data calibration
6	Single-run scenarios (no multi-seed replications)	Run-to-run stochastic variability from vehicle assignment and the Krauss and CACC models was not quantified; part of the observed transition cannot be separated from stochastic noise	Repeat each scenario with multiple random seeds and report variance and confidence intervals
7	No external validation	The model is not validated against observed Beijing volumes or speeds; its outputs should be interpreted as relative, model-based indications of the direction and structure of AV effects rather than as validated predictions of observed conditions	Validate against field data using GEH, RMSE, and MAPE statistics
8	Private commuting vehicles only (no pedestrians, cyclists, buses, or freight)	Non-motorized modes, public transit, and freight are omitted, which could affect the magnitude of the estimated effects, particularly on arterials and in mixed-use corridors	Include multimodal and freight traffic in the simulation

7. Conclusions

Using the SUMO microscopic traffic simulation platform and the complete urban road network of Beijing, this study constructed eleven autonomous vehicle penetration scenarios to systematically evaluate the impacts of autonomous vehicle deployment on city-wide commuting efficiency and the spatial asymmetry of commuting-efficiency gains across the road hierarchy (residential spatial structure is not modeled in this study and is no longer claimed as an outcome). The main conclusions are summarized as follows.

(1) A gradual efficiency transition, occurring over an approximate 30% to 50% penetration range rather than at a single precise threshold, characterizes the realization of autonomous vehicle benefits. Commuting efficiency exhibits a pronounced nonlinear response to autonomous vehicle penetration. Within roughly the lower half of this range, behavioral heterogeneity between autonomous vehicles and human-driven vehicles limits efficiency gains to less than 10%. Beyond this threshold, Cooperative Adaptive Cruise Control enables more stable cooperative traffic flow, resulting in a substantial improvement in network performance. This finding highlights the importance of achieving a critical level of autonomous vehicle adoption to realize system-level benefits.

(2) Autonomous vehicles substantially improve city-wide commuting efficiency under fixed travel demand. Under full autonomous vehicle penetration, the average travel speed increases from 7.20 to 8.27 m/s, while the average simulated travel time decreases from 561 to 270 s, corresponding to improvements of approximately 15% and 52%, respectively. Because these results are obtained under fixed commuting demand, they represent an upper-bound estimate of the operational benefits of autonomous vehicle deployment, and realized long-term benefits may be smaller once induced demand is considered.

(3) Autonomous vehicle benefits exhibit pronounced spatial asymmetry. Efficiency improvements vary systematically across the road hierarchy, reaching 16–20% on expressways and radial commuting corridors, but remaining limited to 4–8% on urban arterial roads with dense signal control. This asymmetry indicates that autonomous vehicle deployment reinforces the operational advantages of higher-order roads rather than producing spatially uniform efficiency gains, reflecting the dependence of cooperative driving on continuous and low-conflict traffic environments.

Future research should incorporate elastic travel demand, vehicle-to-everything infrastructure, and integrated land-use transport models to evaluate the long-term interactions among autonomous vehicle deployment, travel behavior, and urban spatial development. Cross-city comparative studies would further help assess the generalizability of the proposed framework under different road network structures and urban development contexts.

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Data Availability Statement: The SUMO simulation inputs (the network file, vehicle-type definitions with the car-following parameters, detector definitions, and scenario-generation and analysis scripts) that support the findings of this study are available from the corresponding author upon reasonable request. The OpenStreetMap road network data are openly available from https://www.openstreetmap.org (accessed on 13 May 2025).

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